

Search platform user documentation

In this section, we show the user interface of our search platform and explain how to use the application.

Because the majority of datasets in the NKOD contain metadata in Czech, we translated all page content to English using the *Translate to English* feature in Google Chrome.

Searching

To search for datasets, we need to enter a search query into the search bar on the home page and submit it by pressing the search button or by pressing Enter. The search query can be a phrase, a set of keywords, or a full sentence.

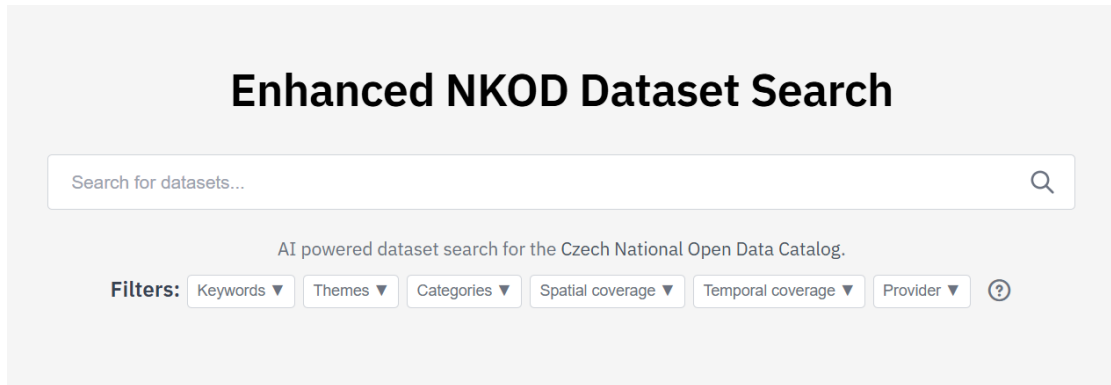


Figure 1: Home page

If we want to apply metadata filters, we can select them in the filters section. We can click on the desired metadata type and check the checkbox next to the desired values. The values are sorted by their number of occurrences. All selected filters will be submitted together with the search query.

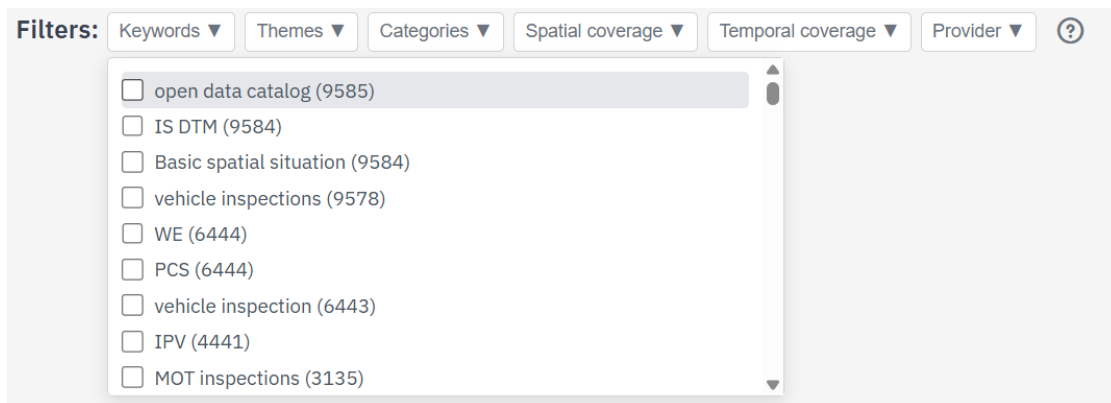


Figure 2: Filters on the home page

The system will start the search based on the submitted query and filters. Once the search results are ready, the search results page will appear. In the following examples, we will use the search results page with the search query *air quality*.

In the center of the search results page is the list of retrieved search results ordered by their relevance. This list is scrollable.

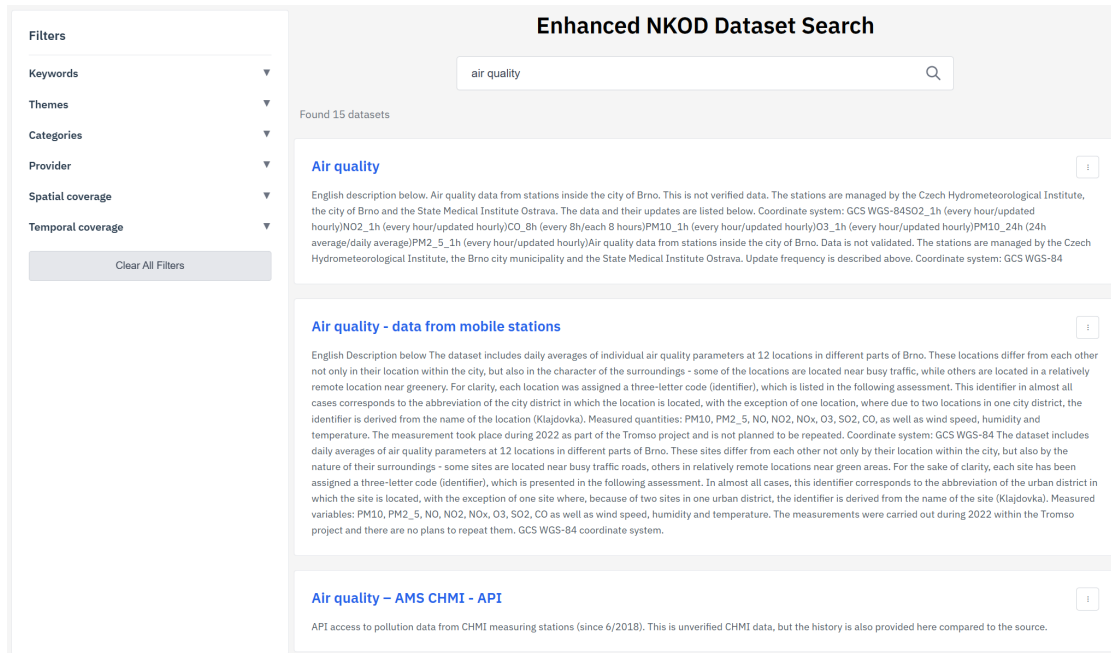


Figure 3: Search results

For each dataset, we show its title and description. To see other metadata, the user can hover over the three dots in the top right corner of the dataset, and a tooltip with the remaining metadata will appear. Each search result has a clickable title, which leads to its dataset detail page.

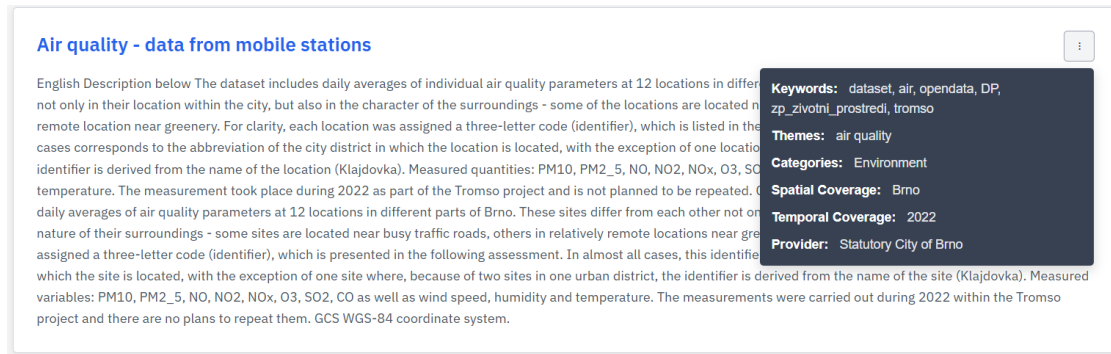


Figure 4: A search result detail

On the left are the metadata filters. Unlike on the home page, here we can select only those metadata that are present in at least one of the results. We can select or deselect the desired metadata using the checkbox next to the value. The filters will be applied immediately to the current results, no reload is needed. To remove all filters, we can use the *Clear all filters* button. In Figure 5, we can see that only two datasets satisfy our selected filters.

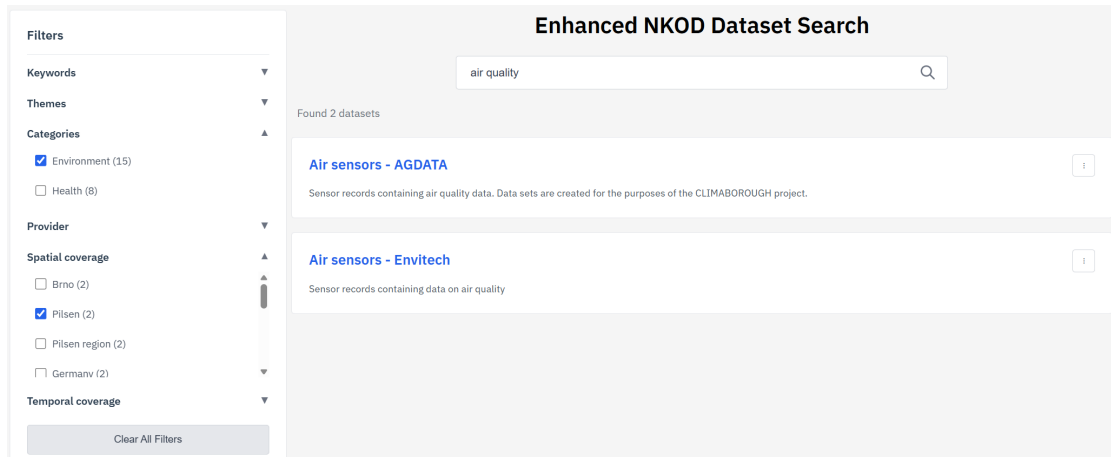


Figure 5: Filtered search results

To perform another search, we can use the search bar at the top, same as on the home page. All selected metadata filters will be passed along with the search query.

Metadata filters

Metadata filters are submitted together with the search query, but they are not strictly enforced. Instead, they are incorporated into the search query. The search is performed, and the selected filters that are present in at least one result are applied to the results. Therefore, selecting a filter and performing a search does not guarantee that all returned results will contain the filter.

On the search results page, we can select metadata filters for the returned results. These filters will be applied immediately as strict constraints on the current results. Immediately after changing the filters, only those results that satisfy all filters will be shown. This can be used to filter the search results so that we do not have to go through all the results by hand.

Empty results

It is possible that no results satisfy the current search query and filters. If this happens, we can try to adjust the metadata filters. We can clear all filters and see if any results were filtered out before. If there are still no results, we need to adjust the search query.

Dataset detail and similar datasets

If we find a dataset that we want to explore more thoroughly, we can click on its title to access its dataset detail page.

Enhanced NKOD Dataset Search

View on the Czech Data Portal

Climate credit in terms of air pollution

Climate credit map of the capital city of Prague - update 2008. The data was created using ArcGIS 9.2, Spatial Analyst. The layer was converted from the credit raster layer, with a horizontal resolution of 25m. The following data were used to create this map: Digital reference map of Prague - block map of buildings Line layer of street sections Vector data of the thematic layer Úpn - traffic - line layer of the road network Outputs from the Model assessment of air quality in Prague - update 2006 by ATEM sro Digital terrain model DMR25 Orthophotos of the Czech Republic. The layer was created by combining the distribution of NO2 and PM10 concentration fields. The resulting layer was reclassified into five categories (I - best for credit quality, V - worst)

Keywords

Air Air pollution Air quality

Categories

Environment

Spatial Coverage

capital city of Prague

Temporal Coverage

2008

Provider

Institute of Planning and Development of the Capital City of Prague

Distributions

ArcGIS Hub Dataset

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ArcGIS GeoService

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CSV

Download file

ZIP

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GeoJSON

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KML

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ZIP

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Figure 6: Dataset detail page

On this page, we can see the dataset metadata. At the top is the dataset title, followed by the description text. Below the description is a section containing keywords, themes, categories, spatial coverage, temporal coverage, and the provider. In Figure 6, the displayed dataset does not have any associated themes, so this metadata type is omitted from the dataset detail.

In the top right corner, there is a button *View on the Czech Data Portal*, which leads to the dataset detail page of this dataset on the Czech Data Portal.

Underneath the provider, the available distributions are displayed. In our example, there are seven distributions. To download the dataset, we can click on *Download file* inside the distribution box.

Similar datasets

Similar datasets are provided at the bottom of the dataset detail page. They are divided into sections based on the type of similarity they show: description, keywords, themes, spatial coverage, temporal coverage, and provider.

Description similarity is based on the semantic similarity between textual representations of the complete datasets' metadata. For keywords, themes, spatial coverage, and temporal coverage, similarity is based on the number of shared metadata of the respective metadata type. If multiple datasets share the same number of metadata values, datasets sharing the highest number of metadata values across the other metadata types are preferred. For provider similarity, we select datasets from the same provider, and we find those that share the highest number of metadata values across all remaining metadata types.

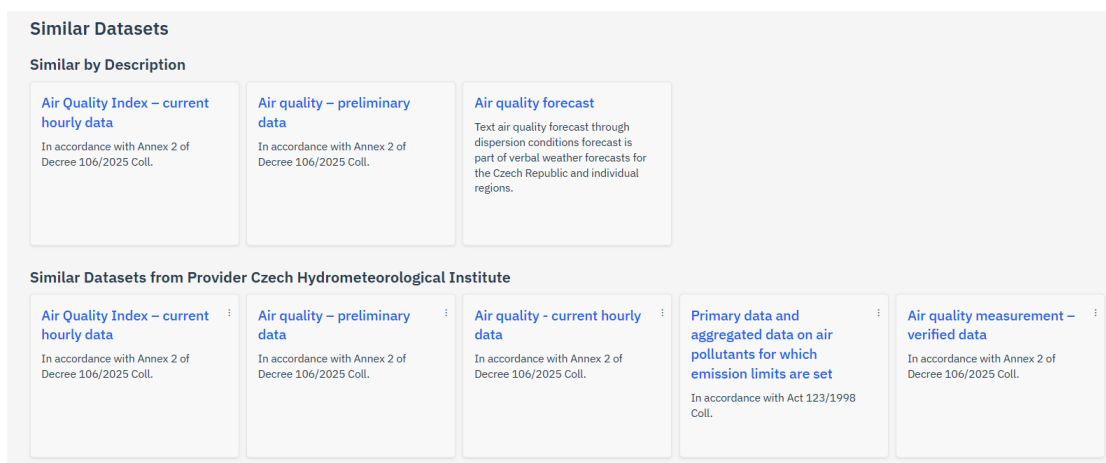


Figure 7: Similar datasets

The same datasets can be marked as similar in different similarity sections. This can be seen in Figure 7, where the first two datasets in both displayed sections are the same.

If the viewed dataset has no similar datasets for a specific similarity type, the corresponding section is not displayed.

For all similarity types except description similarity, we can also see the shared metadata values. To see shared metadata, we can hover over the three dots at the top right corner of the similar dataset box, as is depicted in Figure 8. For provider similarity, shared metadata values across all metadata types are displayed. For the remaining similarity types, we can see shared metadata for the metadata type that the similarity is based on.

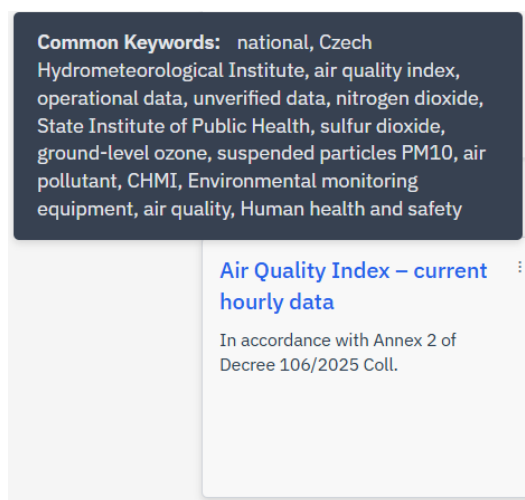


Figure 8: Similar datasets – shared keywords

Chatbot

The chatbot page is accessible from all other pages by clicking on the blue chatbot icon on the right side of the screen. The chatbot interface consists of the chat window, where we can see the chat history. At the bottom, there is the input area where we can write our message. To submit a message, we can press Enter or use the submit button.

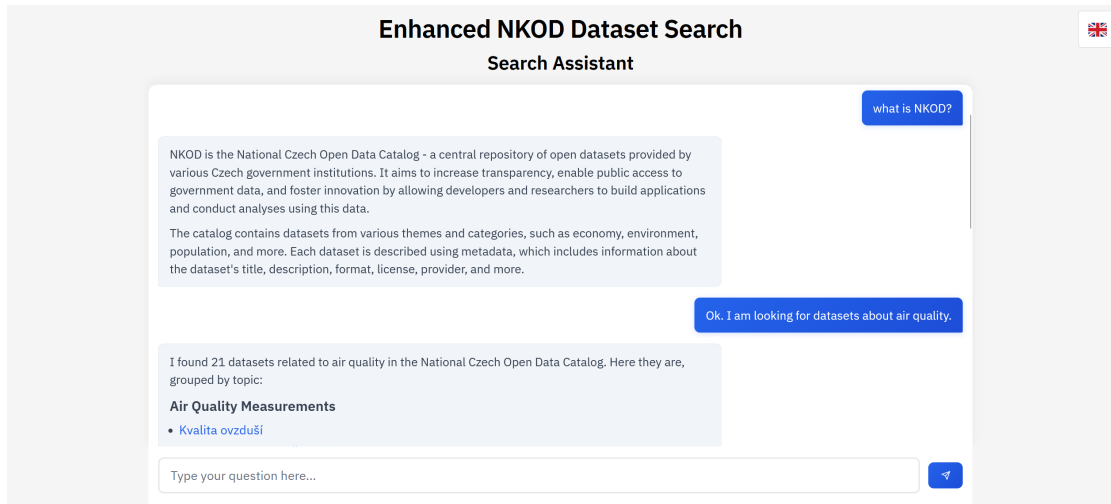


Figure 9: The chatbot page

Language

We can set the chatbot language by clicking on the corresponding flag or language code in the top right corner. The default language is English, and the other supported language is Czech.

Unfortunately, the language setting does not work reliably. The LLM sometimes answers in the wrong language. If we want the response in another language, we can send a follow-up message stating that we want the last chatbot response translated.

Capabilities

The most important functionality of the chatbot is searching in the NKOD. The chatbot starts searching when it detects a need for a search based on the conversation. Then it will construct a message listing the relevant datasets. We can try to refine the search if we are not satisfied with the results, or ask additional questions about the results.

Apart from searching, the chatbot has access to the dataset metadata based on the dataset title. It can also answer simple questions about the NKOD.

Chatbot session

The chatbot session is reset every time the chatbot page is loaded, including reloads. We recommend reloading the page after finishing a conversation to ensure that previous messages will not affect subsequent interactions.